

Transform each of 55-58 by making the change of variable $j = i - 1$.

5.1.57.
$$\sum_{i=1}^{n-1} \frac{i}{(n-i)^2}$$

We must transform the expression by change of variable $j = i - 1$

$$\begin{aligned} j &= i - 1 \\ i &= j + 1 \end{aligned}$$

$$\begin{aligned} \sum_{i=1}^{n-1} \frac{i}{(n-i)^2} &= \\ \sum_{i=1}^{n-1} \frac{j+1}{(n-(j+1))^2} &= \\ \sum_{i=1}^{n-1} \frac{j+1}{(n-j-1)^2} \end{aligned}$$

Now we must also change the lower- and upper bounds.

$$\begin{aligned} i &= j + 1 \\ j + 1 &= 1 \\ j &= 0 \\ j + 1 &= n - 1 \\ j &= n - 2 \end{aligned}$$

Solution:
$$\sum_{j=0}^{n-2} \frac{j+1}{(n-j-1)^2}$$